

FORMULAS & METHODOLOGY

# Referral Program CAC Tracker

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A field-by-field guide to every formula in  
Referral\_Program\_CAC\_Tracker.xlsx

*Every formula below is copied verbatim from the workbook, using row 8 (Install #1) as the worked  
reference row. The same formula pattern repeats down to row 37 (Install #30).*

## 1. Overview

Referral\_Program\_CAC\_Tracker.xlsx is a two-sheet workbook for tracking customer acquisition cost (CAC) across the first 30 installs of the referral program. **Install Log** holds one row per install, with manual-entry deal data and 12 auto-calculating columns. **Summary Dashboard** rolls those 30 rows up into blended totals, a trend check, and a single traffic-light verdict on whether the program is safe to scale.

This document explains what every formula does and why it's built the way it is. It does not replace the workbook — open Referral\_Program\_CAC\_Tracker.xlsx to enter data; use this guide as the reference when a formula needs checking or editing.

## 2. Editable Assumptions

Three cells at the top of Install Log drive every row's commission and discount math. They're the only inputs meant to change after the program launches.

| Cell      | Assumption                  | Current Value | Notes                                                                                                                                                                          |
|-----------|-----------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>D3</b> | Commission Rate (all tiers) | 5.0%          | Applied to every install regardless of tier, per the current program structure. Referenced by every row's Commission % formula — change it once here and all 30 rows re-price. |
| <b>D4</b> | Discount Rate – Above 250k  | 4.0%          | Discount applied to installs valued at KES 250,000 or more.                                                                                                                    |
| <b>D5</b> | Discount Rate – 150k–250k   | 3.0%          | Discount applied to installs valued between KES 150,000 and KES 249,999.                                                                                                       |

## 3. Install Log – Column Reference

Columns A–E and K, N are manual entry (blue text in the workbook). Every other column is a formula (black text) and recalculates automatically as soon as the manual cells in its row are filled in.

### Deal Basics

#### **A** Install #

FORMULA

```
=ROW() - 7
```

Sequential install number, derived from the row position so it stays correct even if rows are reordered. Row 8 is Install #1, row 37 is Install #30.

#### **B** Referral Code

MANUAL ENTRY

The unique code the farmer quoted when they first contacted G&P.

**C Referrer Name**

MANUAL ENTRY

Who the commission and program credit belong to.

**D Installation Date**

MANUAL ENTRY

Date the installation was completed and the deal closed.

**E Installation Value (KES)**

MANUAL ENTRY

The closed deal value. Every downstream formula in the row keys off this cell — leaving it blank keeps the whole row blank.

**F Tier**

FORMULA

```
=IF(E8="", "", IF(E8>=250000, "Above 250k", IF(E8>=150000, "150k-250k", "Below 150k")))
```

Buckets the install by Installation Value. "Below 150k" is a safety net the spec didn't define a rate for — see the note under Discount % below.

**Commission****G Commission %**

FORMULA

```
=IF(E8="", "", $D$3)
```

Pulls the flat rate from the editable Commission Rate cell (D3) rather than hardcoding 5% into every row.

**H Commission Paid (KES)**

FORMULA

```
=IF(E8="", "", E8*G8)
```

Installation Value × Commission %.

**Discount****I Discount %**

FORMULA

```
=IF(F8="", "", IF(F8="Above 250k", $D$4, IF(F8="150k-250k", $D$5, 0)))
```

Looks up the rate for the row's Tier from D4/D5. A "Below 150k" install gets 0% here — the program terms didn't specify a rate for that band, so it defaults to no discount rather than guessing at one.

**J Discount Given (KES)**

FORMULA

```
=IF(E8="", "", E8*I8)
```

Installation Value × Discount %.

## Margin

### **K** Cost of Materials/Labor/Logistics (KES)

MANUAL ENTRY

All-in delivered cost of the install: panels, inverter, labor, logistics.

### **L** Gross Margin (KES)

FORMULA

```
=IF(OR(E8="",K8=""), "", E8-K8)
```

Installation Value – Cost. Blank until both E and K are filled in, so an unentered cost never silently reads as a zero-cost, 100%-margin install.

### **M** Gross Margin %

FORMULA

```
=IF(OR(L8="",E8="",E8=0), "", L8/E8)
```

Gross Margin ÷ Installation Value.

## Acquisition Cost & Risk

### **N** Overhead Allocation (KES)

MANUAL ENTRY

This install's share of landing-page cost, admin time, and other program overhead — however you choose to divide it across installs.

### **O** Total Acquisition Cost (KES)

FORMULA

```
=IF(OR(E8="",N8=""), "", H8+J8+N8)
```

Commission Paid + Discount Given + Overhead Allocation.

### **P** CAC % of Deal Value

FORMULA

```
=IF(OR(O8="",E8="",E8=0), "", O8/E8)
```

Total Acquisition Cost ÷ Installation Value.

### **Q** CAC-to-Margin Ratio

FORMULA

```
=IF(OR(O8="",L8="",L8=0), "", O8/L8)
```

Total Acquisition Cost ÷ Gross Margin — the core health metric: how much of the profit on this install got spent acquiring it.

**R****Flag**

FORMULA

```
=IF(Q8="", "", IF(Q8>0.4, "High Risk", IF(Q8>=0.3, "Caution", "Healthy")))
```

Labels the row from its CAC-to-Margin Ratio. Conditional formatting colors this cell red above 0.40, yellow from 0.30–0.40, and green below 0.30.

## 4. Summary Dashboard — Metric Reference

Every metric below reads from the 30-row range Install Log!8:37, so it updates automatically as rows are filled in — no manual refresh needed.

### Totals

**B3****Installs Logged**

```
=COUNT('Install Log'!E8:E37)
```

Counts rows with an Installation Value entered, out of the 30 available.

**B4****Total Installation Value (KES)**

```
=SUM('Install Log'!E8:E37)
```

**B5****Total Commissions Paid (KES)**

```
=SUM('Install Log'!H8:H37)
```

**B6****Total Discounts Given (KES)**

```
=SUM('Install Log'!J8:J37)
```

**B7****Total Overhead Allocated (KES)**

```
=SUM('Install Log'!N8:N37)
```

## Blended CAC Metrics

### B10 Average Deal Value (KES)

```
=IF(COUNT('Install Log'!E8:E37)=0,"",SUM('Install Log'!E8:E37)/COUNT('Install Log'!E8:E37))
```

Total Installation Value ÷ Installs Logged.

### B11 Blended CAC (KES)

```
=IF(COUNT('Install Log'!E8:E37)=0,"",SUM('Install Log'!O8:O37)/COUNT('Install Log'!E8:E37))
```

Total Acquisition Cost across all logged installs ÷ Installs Logged — the average acquisition cost per install.

### B12 Blended CAC % of Avg Deal Value

```
=IF(OR(B10="",B10=0),"",B11/B10)
```

Blended CAC ÷ Average Deal Value.

### B13 Average Gross Margin %

```
=IFERROR(AVERAGE('Install Log'!M8:M37),"")
```

Straight average of each logged row's Gross Margin %.

### B14 Average CAC-to-Margin Ratio

```
=IFERROR(AVERAGE('Install Log'!Q8:Q37),"")
```

Straight average of each logged row's CAC-to-Margin Ratio. This is the number the Program Status box (below) is built on.

## Trend Check — Installs 1–15 vs 16–30

### B17 Avg CAC-to-Margin Ratio, Installs 1–15

```
=IFERROR(AVERAGE('Install Log'!Q8:Q22),"")
```

### B18 Avg CAC-to-Margin Ratio, Installs 16–30

```
=IFERROR(AVERAGE('Install Log'!Q23:Q37),"")
```

B19

Trend Indicator

=IF(OR(B17="",B18=""),"Insufficient Data",IF(B18>B17\*1.05,"Climbing",IF(B18<B17\*0.95,"Improving","Stable")))

"Climbing" if the second-half average is more than 5% above the first half; "Improving" if more than 5% below; otherwise "Stable". The 5% band is an assumption, not a figure from the program terms — the 1.05 / 0.95 multipliers in the formula are what to edit to change the sensitivity.

## 5. Program Status Logic

The large status cell on the dashboard evaluates this formula against Average CAC-to-Margin Ratio (B14) and the Trend Indicator (B19):

=IF(B14="", "INSUFFICIENT DATA", IF(B14>0.4, "RENEGOTIATE TERMS", IF(B14>=0.35, "REVIEW NEEDED", IF(B19="Climbing", "REVIEW NEEDED", "SAFE TO SCALE"))))

| Result            | Trigger condition                                                                                             |
|-------------------|---------------------------------------------------------------------------------------------------------------|
| RENEGOTIATE TERMS | Average CAC-to-Margin Ratio (B14) is above 0.40.                                                              |
| REVIEW NEEDED     | Average CAC-to-Margin Ratio is 0.35–0.40 — or it's under 0.35 but the Trend Indicator (B19) reads "Climbing." |
| SAFE TO SCALE     | Average CAC-to-Margin Ratio is under 0.35 and the trend is not "Climbing."                                    |

Reading note: a climbing trend keeps the status out of "SAFE TO SCALE" even when the average ratio is still under 0.35 — it falls back to "REVIEW NEEDED" instead, since a rising CAC trend isn't "stable" even if it hasn't crossed a hard threshold yet. This is a judgment call built into the nested IF above, not a number from the original program terms — adjust the formula directly if a different interpretation is wanted.

## 6. Worked Example — Install #1

Row 8 in the shipped workbook is a filled-in example so the expected format is obvious. Here is the same row traced through every formula by hand:

|                                            |                                                                          |
|--------------------------------------------|--------------------------------------------------------------------------|
| Installation Value (KES)                   | <b>275,000</b>                                                           |
| Tier                                       | <b>Above 250k (<math>\geq 250,000</math>)</b>                            |
| Commission % $\rightarrow$ Commission Paid | <b><math>5.0\% \times 275,000 = \text{KES } 13,750</math></b>            |
| Discount % $\rightarrow$ Discount Given    | <b><math>4.0\% \times 275,000 = \text{KES } 11,000</math></b>            |
| Cost of Materials/Labor/Logistics (KES)    | <b>190,000</b>                                                           |
| Gross Margin $\rightarrow$ Gross Margin %  | <b><math>275,000 - 190,000 = \text{KES } 85,000</math> (30.9%)</b>       |
| Overhead Allocation (KES)                  | <b>3,000</b>                                                             |
| Total Acquisition Cost                     | <b><math>13,750 + 11,000 + 3,000 = \text{KES } 27,750</math></b>         |
| CAC % of Deal Value                        | <b><math>27,750 \div 275,000 = 10.1\%</math></b>                         |
| CAC-to-Margin Ratio $\rightarrow$ Flag     | <b><math>27,750 \div 85,000 = 0.33 \rightarrow \text{Caution}</math></b> |